HW2: Logistic Regression

STAT 3302

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Problem 1: When linear regression goes wrong.

a) Unbiasedness of the OLS estimator

1. Compute $\mathbb{E}[Y_i|X_i]$.

$$\mathbb{E}(Y_i|X_i) = P(Y_i = 1|X_i) = \frac{e^{\beta_0+\beta_1 X_i}}{1+e^{\beta_0+\beta_1 X_i}}$$

2. Show that the OLS estimator $\hat{\beta}_1$ is unbiased for β_1.

The OLS estimator $\hat{\beta}_1$ is unbiased because the conditional mean of Y given X is: $\mathbb{E}[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$. Using the OLS slope formula $\hat{\beta}_1 = \frac{\sum_i (X_i-\bar{X})(Y_i-\bar{Y})}{\sum_i (X_i-\bar{X})^2}$ and taking expectations conditional on X gives $\mathbb{E}[\hat{\beta}_1 \mid X] = \frac{\sum_i (X_i-\bar{X})(\mathbb{E}[Y_i|X_i]-\overline{\mathbb{E}[Y|X]})}{\sum_i (X_i-\bar{X})^2} = \beta_1$. Therefore $\hat{\beta}_1$ is unbiased.

b) Conditional variance of the errors

Define the regression error as $\epsilon_i = Y_i - (\beta_0 + \beta_1 X_i)$.

1. Compute Var($Y_i|X_i$).

$\mathrm{Var}(Y_i|X_i) = p(1-p) =$

$$\frac{e^{\beta_0+\beta_1 X_i}}{1+e^{\beta_0+\beta_1 X_i}}(1 - \frac{e^{\beta_0+\beta_1 X_i}}{1+e^{\beta_0+\beta_1 X_i}}) = \frac{e^{\beta_0+\beta_1 X_i}}{(1+e^{\beta_0+\beta_1 X_i})^2}$$

2. Deduce Var($\epsilon_i|X_i$).

$$Var(\epsilon_i|X_i) = Var(Y_i - \mathbb{E}(Y_i|X_i)|X_i) = Var(Y_i|X_i)$$

$\therefore Var(\epsilon_i|X_i) = \frac{e^{\beta_0+\beta_1 X_i}}{1+e^{\beta_0+\beta_1 X_i}}$

3. Is the assumption Var$(\epsilon_i|X_i) = \sigma^2$ satisfied?

No, it is not. Our error is larger when p=0.50 compared to near 0 or 1, which violates homoskedasticity. OLS is still unbiased, but inference is invalid.

c) Reported confidence interval

Alex reported a 95% confidence interval for β_1 using standard OLS formula.

1. Identify which assumption(s) required for this confidence interval fail in this setting.

- The errors are heteroskedastic as proven above.
- The constant variance assumption also fails as variance depends on X_i.
- And of course, we also cannot make the assumption of independent, normal errors.

2. Is this confidence interval valid?

No, this confidence interval is not valid. This is because our errors are heteroskedastic, the standard OLS formula underestimates the true variability, so the reported 95% confidence interval will not achieve the correct coverage probability.

d) Communication with the client

Alex tells the client: "The effect of X on the probability of success is statistically significant at the 5% level." Based on your analysis, is this statement justified? What is the main statistical mistake Alex made?

Alex's statement is not justified. although the point estimate is unbiased, the confidence interval and p-value are invalid due to heteroskedastic and not normally distributed errors. Alex's main mistake was assuming the standard OLS formulas for inference are valid for a linear probability model, ignoring that $Var(\epsilon_i|X_i)$ depends on X_i. Thus, his statement about statistical significance at the 0.05 level is unreliable.

e) Reflection (short answer)

In 2–3 sentences, explain **why generalized linear models** are better suited than linear regression for binary response data.

Generalized linear models are better suited for this because they use a link function which contains the predicted probabilities between 0 and 1. They also account for heteroskedasticity where variance depends on the mean. In contrast, OLMs assume constant variance and can provide nonsensical predictions or standard errors when applied to binary data.

Problem 2: Enter Generalized Linear Models (GLMs)

a) Exponential family: the generic form

A random variable $Y is said to belong to the exponential family if its probability mass function (or density) can be written as

$$f(y|\theta, \phi) = exp(\frac{y\theta - b\theta}{a\phi} + c(y, \phi)),$$

`instructed to ignore...`

b) Mean structure and link function

Show that the expectation has the form $b'(\theta)$:

The log-likelihood for a single observation is:

$$\updownarrow(\theta; y) = \frac{y\theta - b\theta}{a\phi} + c(y, \phi)$$

The derivative is:

$$\frac{\partial l}{\partial \theta} = \frac{y - b'(\theta)}{a(\phi)}$$

Thus:

$$\mathbb{E}[\frac{\partial l}{\partial \theta}] = 0 \implies \mathbb{E}[\frac{y - b'(\theta)}{a(\phi)}] = 0 \qquad \therefore \mathbb{E}[y] = b'(\theta)$$

c) Bernoulli response -> logistic regression

1. Show that the Bernoulli distribution can be written in exponential form.

The Bernoulli pmf is $P(Y = y) = p^y(1-p)^{1-y}$. The log of that is:

$$e^{y\log(p)+(1-y)\log(1-p)} = e^{y\log\frac{p}{1-p}+\log(1-p)}$$

If we compare $y\log\frac{p}{1-p} + \log(1-p)$ with $y\theta - b(\theta)$, the only θ that fits is $\log\frac{p}{1-p}$, which would make $\log(1-p) = -b(\theta)$. $b(\theta) = -\log(1-p) \implies b(\theta) = \frac{1}{1+e^\theta} = \log(1+e^\theta)$.

Thus the Bernoulli distribution expressed in the exponential form is $exp(y\theta - \log(1+e^\theta))$.

2. Identify the sufficient statistic and natural parameter θ.

As shown above, the natural parameter is $\theta = \log(\frac{p}{1-p})$. This makes the sufficient statistic y=y itself, since $y\log\frac{p}{1-p} + \log(1-p) = y\theta/a(\phi) - b\theta/a(\phi)$. The dispersion $a(\phi)$ is also 1.

3. Show that $\theta = \log(\frac{p}{1-p})$.

$$
\begin{aligned}
P(Y = y) &= p^y(1-p)^{1-y} \\
\log P(Y = y) &= y \log p + (1-y)\log(1-p) \\
((1-y)\log(1-p) &= \log(1-p) - y\log(1-p)) \\
\text{By log subtraction property, } \log P(Y = y) &= y \log \frac{p}{1-p} + \log(1-p) \\
P(Y = y) &= exp(y \log \frac{p}{1-p} + \log(1-p))
\end{aligned}
$$

The exponential family has a structure $f(y, \theta) = exp(y\theta - b(\theta))$. Comparing, the coefficient of y **must be** the natural parameter, and the coefficient of y is $\log \frac{p}{1-p}$.

d) Normal response -> Linear regression

i. Write the normal density in exponential family form. Clearly identify: - the sufficient statistic - the natural (canonical) parameter - the dispersion parameter.

Normal density is:

$$
\begin{aligned}
f(y; \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(y-\mu)^2}{2\sigma^2}) \\
(y-\mu)^2 &= y^2 - 2y\mu + \mu^2 \\
\text{So: } f(y) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{y^2}{2\sigma^2} + \frac{uy}{\sigma^2} + \frac{\mu^2}{2\sigma^2}) \\
\text{General exponential family form: } & \exp(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi))
\end{aligned}
$$

Matching terms:

$$
f(y) = \exp(\frac{y\mu - 0.5\mu^2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2))
$$

Identifying components, we see:

- Sufficient statistic is y.
- Natural parameter is $\theta = \mu$.
- Dispersion parameter is $a(\phi) = \sigma^2$.

ii. Using the exponential family representation, derive an expression for $\mathbb{E}(Y_i)$ in terms of the natural parameter.

We can see that $b(\theta) = 0.5\theta^2$. Taking the derivative, $b'(\theta) = \theta$. Therefore, $\mathbb{E}(Y_i) = b'(\theta) = \theta$.

iii. Determine the canonical link function for the normal distribution and show that it corresponds to $\mu_i = x_i^\top \beta$.

We know that $\mu = \theta$ from above. Since the canonical link sets $\eta_i = \theta_i$, and we have $\theta_i = \mu_i$, it follows that $\eta_i = \mu_i$.

Therefore the canonical link $g(\mu) = \mu$. It is its own link function.

This corresponds to linear regression, because in GLM:

$$g(\mu_i) = x_i^\top \beta$$

With the identity link:

$$\mu_i = x_i^\top \beta$$

Since $\mu_i = (Y_i)$, we have shown that $\mathbb{E}(Y_i) = x_i^\top \beta$.

iv. Explain why ordinary least squares linear regression can be viewed as a special case of a generalized linear model. Your explanation should explicitly reference the response distribution, the link function, and the role of the linear predictor.

1. In GLM we assume each response Y_i comes from an exponential-family distribution. In OLS linear regression, we assume $Y_i|x_i \sim \mathcal{N}(\mu_i, \sigma^2)$. The normal distribution **is** a member of the exponential family distribution as we had shown above, so it is admissable by GLM.
2. We had also shown that the canonical link is itself: $g(\mu) = \mu$.
3. The linear predictor is a combination of covariates and coefficients. In OLS it is the same object, $x_i^\top \beta$.
4. MLE under these assumptions coincides with least squares.

OLS is just GLM with Normal errors and identity link.

e) Poisson response -> Log-linear regression

i. Write the Poisson pmf in the exponential family form. Identify the sufficient statistic, natural parameter, and dispersion parameter.

$$\begin{aligned} P(Y = y) &= \frac{\lambda^y e^{-\lambda}}{y!} \\ \log P(Y = y) &= \exp(y \log \lambda - \lambda - \log(y!)) \\ Matching: & \\ f(y; \theta, \phi) &= \exp(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)) \end{aligned}$$

Comparing term by term:

- $\theta = \log \lambda$.
- The sufficient statistic is $y = y$.
- The function $b(\theta) = e^\theta = \lambda$.
- The dispersion parameter is $a(\phi) = 1$.
- Additionally, $c(y, \phi) = -\log(y!)$.

ii. Show that $\theta = \log(\lambda)$.

$$\log P(Y = y) = \exp(y \log \lambda - \lambda - \log(y!))$$

Since θ must be the coefficient of y, $\theta = \log \lambda$.

iii. Determine the canonical link function and show that it leads to the log-linear model $\log(\lambda_i) = x_i^\top \beta$.

We know $b'(\theta) = e^\theta$. Thus $\mu = \mathbb{E}(Y) = e^\theta$. But for the Poisson distribution, $\mu = \lambda$. Therefore we obtain $\lambda = e^\theta$, which means $\theta = \log \lambda$.

The canonical link sets $\eta_i = \theta_i$. Since $\theta_i = \log \lambda_i$, the canonical link is $g(\mu) = \log \mu$, because $\mu = \lambda$ for Poisson. This is the log link.

In a GLM, $g(\mu_i) = \eta_i = x_i^\top \beta$, but using the canonical link, $\log(\lambda_i) = x_i^\top \beta$. This is exactly the **log-linear model**.

iv. Explain why Poisson is often referred to as a log-linear model. Your explanation should reference the response distribution, the link function, and how covariates affect the mean of Y_i.

The Poisson is referred to as the **log-linear model** because: - The response distribution is Poisson: $\mathbb{E}(Y_i|x_i) = \lambda_i$. - The canonical link is the log link: $\log(\lambda_i) = \eta_i$. - The linear predictor models $\log(\lambda_i)$, making the log-mean a function of the covariates. - Covariates have multiplicative effects on the expected count. For instance, a one-unit increase in x_{ij} would change the mean by a factor of $\exp(\beta_j)$.

Problem 3: Logistic Regression on the Titanic Data

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.3.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(titanic)
```

```
## Warning: package 'titanic' was built under R version 4.3.3
```

```r
titanic_raw <- titanic_train
titanic_clean <- titanic_raw %>%
  filter(
    !is.na(Survived),
    !is.na(Age),
    !is.na(Sex)
  )
```

a) Binning age and estimating survival probabilities

1. Create bins for `Age`.

```
titanic_binned <- titanic_clean %>%
  mutate(
    AgeGroup = cut(
      Age,
      breaks = c(0, 12, 18, 30, 45, 60, Inf),
      labels = c("Child", "Teen", "YoungAdult", "Adult", "MiddleAged", "Senior"),
      right = FALSE
    )
  )
```

2. Within each age bin, and separately for each combination of sex and passenger class, compute the proportion of passengers who survived and compute the log-odds (logit) of survival using these proportions.

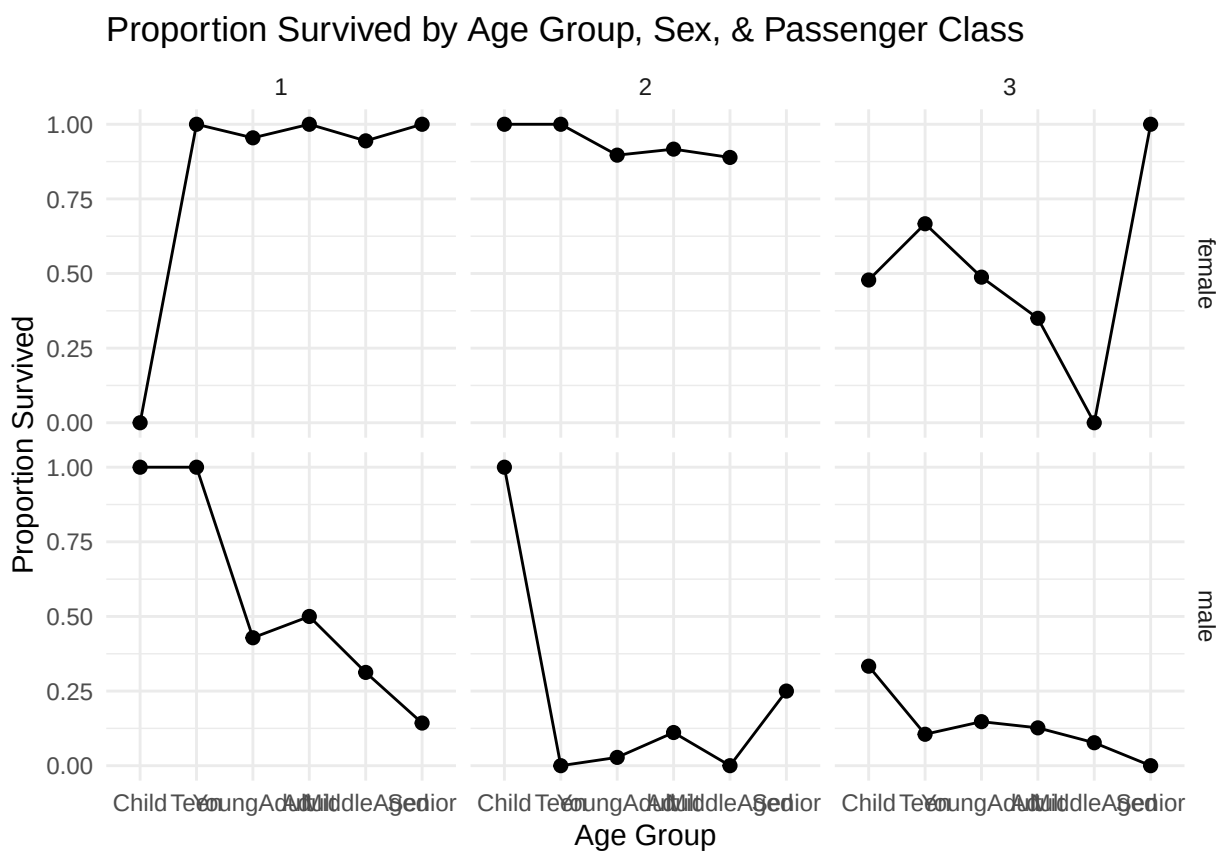
```
survival_by_group <- titanic_binned %>%
  group_by(AgeGroup, Sex, Pclass) %>%
  summarise(
    n = n(),
    survived = sum(Survived == 1),
    prop_survived = survived/n,
    logit_survived = log(prop_survived / (1 - prop_survived)),
    .groups = "drop"
  )
survival_by_group
```

```
## # A tibble: 35 x 7
##    AgeGroup Sex    Pclass     n survived prop_survived logit_survived
##    <fct>    <chr>   <int> <int>    <int>         <dbl>          <dbl>
##  1 Child    female      1     1        0         0          -Inf
##  2 Child    female      2     8        8         1           Inf
##  3 Child    female      3    23       11         0.478        -0.0870
##  4 Child    male        1     3        3         1           Inf
##  5 Child    male        2     9        9         1           Inf
##  6 Child    male        3    24        8         0.333        -0.693
##  7 Teen     female      1     7        7         1           Inf
##  8 Teen     female      2     4        4         1           Inf
##  9 Teen     female      3    12        8         0.667         0.693
## 10 Teen     male        1     1        1         1           Inf
## # i 25 more rows
```

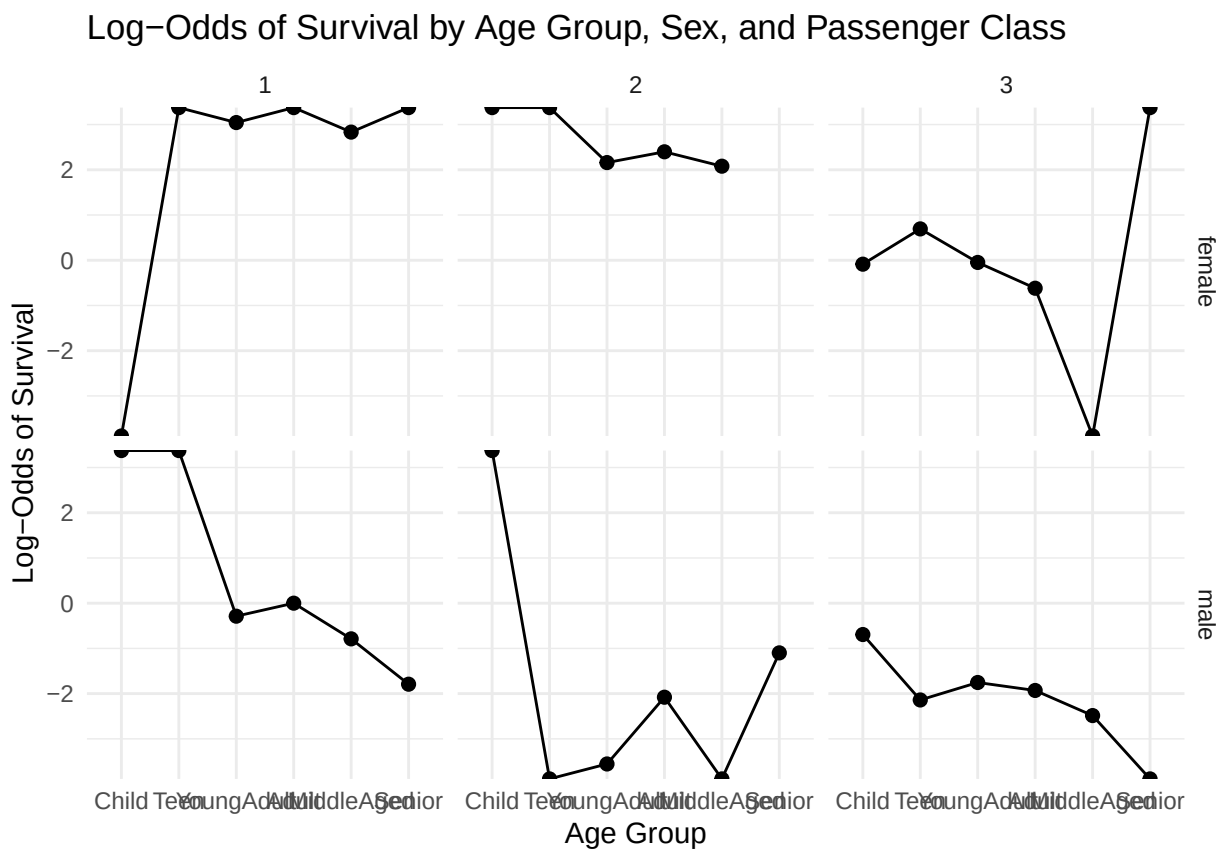
3. Create exploratory plots to visualize: a plot of proportion survived vs age bin, faceted by sex and passenger class and a plot of log-odds of survival vs age bin, faceted by sex and passenger class.

```r
library(ggplot2)

ggplot(survival_by_group,
       aes(x = AgeGroup, y = prop_survived, group=1)) +
  geom_line() +
  geom_point(size = 2) +
  facet_grid(Sex ~ Pclass) +
  labs(
    title="Proportion Survived by Age Group, Sex, & Passenger Class",
    x = "Age Group",
    y = "Proportion Survived") +
  theme_minimal()
```




```r
ggplot(survival_by_group,
       aes(x = AgeGroup, y = logit_survived, group = 1)) +
  geom_line() +
  geom_point(size = 2) +
  facet_grid(Sex ~ Pclass) +
  labs(
    title = "Log-Odds of Survival by Age Group, Sex, and Passenger Class",
    x = "Age Group",
    y = "Log-Odds of Survival"
  ) +
  theme_minimal()
```



b) Association between sex and survival: Fisher's Exact Test

- Null Hypothesis (H_0): Sex and survival are independent.
- Alternative Hypothesis (H_1): They are not.

1. Construct a 2×2 contingency table for `Sex` and `Survived`.

```
tab_sex_survive <- table(
  titanic_clean$Sex,
  titanic_clean$Survived)

tab_sex_survive
```

```
##
##            0   1
##   female  64 197
##   male   360  93
```

2. Perform Fisher's Exact Test to assess whether survival is related to sex.

```r
fisher.test(
  x=tab_sex_survive)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  tab_sex_survive
## p-value < 2.2e-16
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.05738561 0.12249584
## sample estimates:
## odds ratio
##  0.0843102
```

3. Report the p-value of the test and whether to reject the Null at the 5% significance level.

The p-value is reported to be under 2.2e-16 or 0.00000000000000022, very close to 0. This is very significant at the 0.05 level, giving us reason to reject the null hypothesis.

4. Explain what independence means in this context, and what conclusion the test supports regarding sex and survival.

This test is a test on whether the two sexes' survivability is independent. In this context, that means whether sex and survival are statistically associated.

In other words, survival probabilities differ between males and females, and because we rejected the null hypothesis, we can conclude that knowing a passenger's sex does give additional information about their likelihood of surviving.

c) Logistic regression for survival on the Titanic

1. Fit a logistic regression model using `Age`, `Sex`, and `Pclass` as predictors.

```r
titanic_clean <- titanic_clean %>%
  mutate(Pclass=factor(Pclass)) # This enables categories

model <- glm(Survived ~ Age + Sex + Pclass,
             data=titanic_clean,
             family=binomial)
summary(model)
```

```
##
## Call:
## glm(formula = Survived ~ Age + Sex + Pclass, family = binomial,
##     data = titanic_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)  3.777013   0.401123   9.416  < 2e-16 ***
## Age         -0.036985   0.007656  -4.831 1.36e-06 ***
## Sexmale     -2.522781   0.207391 -12.164  < 2e-16 ***
## Pclass2     -1.309799   0.278066  -4.710 2.47e-06 ***
## Pclass3     -2.580625   0.281442  -9.169  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 964.52  on 713  degrees of freedom
## Residual deviance: 647.28  on 709  degrees of freedom
## AIC: 657.28
##
## Number of Fisher Scoring iterations: 5
```

2. What is a reference category? How are dummy (indicator) variables constructed for `Sex` and `Pclass`? Which levels are used as reference levels in your model?

A reference category is a level whose coefficient is set to 0, and it is the baseline group against which all other groups are interpreted.

A dummy variable is constructed as a binary decision. For sex, it is 1 if Sex=male, else 0 (meaning female is the reference). For Pclass there are 3 levels, so 2 dummy variables must be created. Pclass2 is 1 if Pclass=2 else 0, and Pclass3 is 1 if Pclass=3, else 0. Pclass1 is the reference.

3. Determine how many regression coefficients are estimated in the final model and how many of these correspond to categorical predictors.

There are 5 regression coefficients in the fitted model: the intercept, one for Age, one for sex being male, and 2 for whether the Pclass is 2 or 3 (1 is baseline.)

There are 3 categorical predictors. The dummy variable for Sexmale, and the 2 dummy variables for Pclass2 and Pclass3.

4. Interpret each estimated coefficient on the log-odds scale and then on the odds scale. Your interpretation should clearly state which group is being compared to which, and which variables are held fixed.

1. Intercept: 3.777

The expected log-odds of survival for females in 1st class, age 0 is 3.777.

On the odds scale, this is ~43.8; odds of survival are very high for the baseline group (though age 0 is extrapolated).

2. Age: 0.036985

   Note, this is part (5):

Each additional year of age decreases the log-odds of survival slightly, holding Sex and Pclass constant.

On the odds scale this is 0.964. This means each extra year of age reduces the odds of survival by ~3.6%, holding Sex and Pclass fixed.

3. Sexmale: -2.522781

Being male vs female reduces the log-odds of survival by 2.523, holding Pclass and Age constant.

The odds ration of this is 0.080, meaning males have 8% of the odds of survival compared to females of the same age and passenger class.

4. Pclass2: -1.309799

Moving from 1st to 2nd class reduces log-odds of survival by -1.310, holding Age and Sex constant.

With an odds of 0.270, we can say 2nd class passengers have 27% of the odds of survival compared to 1st class passengers of the same sex/age.

5. Pclass3: -2.580625

Moving from 1st class to 3rd class reduces the log-odds of survival by -2.58, holding Age and Sex constant.

Interpreting the odds of 0.076, we can say 3rd class passengers have **7.6% of the odds of survival** compared to 1st class passengers of the same sex and age.

d) Assessing variable importance using likelihood ratio tests

1. Using `anova()` with `test="Chisq"`, test the contributions of Age, Sex, and Pclass by comparing appropriately nested models.

```r
full_model <- glm(Survived ~ Age + Sex + Pclass,
                  data=titanic_clean,
                  family= binomial)

# Nested models
model_no_age   <- glm(Survived ~ Sex + Pclass, data = titanic_clean, family = binomial)
model_no_sex   <- glm(Survived ~ Age + Pclass, data = titanic_clean, family = binomial)
model_no_pclass <- glm(Survived ~ Age + Sex, data = titanic_clean, family = binomial)

"Test contribution of Age"
```

```
## [1] "Test contribution of Age"
```

```r
anova(model_no_age, full_model, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Survived ~ Sex + Pclass
## Model 2: Survived ~ Age + Sex + Pclass
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1       710     672.43
## 2       709     647.28  1   25.148 5.311e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
""
```

```
## [1] ""
```

```r
"Test contribution of Sex"
```

```
## [1] "Test contribution of Sex"
```

```r
anova(model_no_sex, full_model, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Survived ~ Age + Pclass
## Model 2: Survived ~ Age + Sex + Pclass
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1       710     827.16
## 2       709     647.28  1   179.88 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
""
```

```
## [1] ""
```

```r
"Test contribution of Pclass"
```

```
## [1] "Test contribution of Pclass"
```

```r
anova(model_no_pclass, full_model, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Survived ~ Age + Sex
## Model 2: Survived ~ Age + Sex + Pclass
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1       711     749.96
## 2       709     647.28  2   102.67 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2. Using the `drop1()` function, perform a likelihood ratio test for each predictor in the model, and report the p-value associated with dropping each term.

```r
drop1(full_model, test = "Chisq")
```

```
## Single term deletions
##
## Model:
## Survived ~ Age + Sex + Pclass
##        Df Deviance    AIC     LRT  Pr(>Chi)
## <none>      647.28 657.28
## Age     1   672.43 680.43  25.148 5.311e-07 ***
## Sex     1   827.16 835.16 179.881 < 2.2e-16 ***
## Pclass  2   749.96 755.96 102.674 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Pr for Age was 5.311e-7, Pr for Sex was <2.2e-16, and Pr for Pclass was 2.2e-16. These are all extremely significant, proving they are significant contributors to survivability.

3. Explain the difference between testing a single coefficient vs testing a multi-level categorical variable such as `Sex` or `Pclass`.

There are multiple differences:

- The df for a single coefficient is only 1, while for multi-level categorical variables it is the # of non-reference levels.
- The hypothesis for a single coefficient is that one param = 0. For multi-level categorical, it is that all dummy params are = 0.
- The interpretation for one coefficient is the effect of one specific variable, whereas for multi-level categorical it is the overall effect of the category.

4. Perform a joint test to assess whether `Sex` and `Pclass` together contribute to the model beyond `Age`. Clearly state the full model, reduced model, and null hypothesis being tested.

Full Model: Survival ~ Age + Sex + Pclass

Reduced Model: Survival ~ Age

The null hypothesis H_0: $\beta_{Sex} = \beta_{Pclass2} = \beta_{Pclass3} = 0$. The alternative hypothesis H_1: At least one of the coefficients for Sex or Pclass $\neq 0$.

```r
# Age + Sex + Pclass
full_model <- glm(Survived ~ Age + Sex + Pclass,
                  data = titanic_clean,
                  family = binomial)

# Age only
reduced_model <- glm(Survived ~ Age,
                     data = titanic_clean,
                     family = binomial)

anova(reduced_model, full_model, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Survived ~ Age
## Model 2: Survived ~ Age + Sex + Pclass
```

```
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1       712     960.23
## 2       709     647.28  3   312.94 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value of the full model is < 2.2e-16, which is far below 0.05 significance. This means Sex and Pclass together strongly affect survival, even after accounting for Age.

e) Visualizing fitted probabilities and making predictions

1. Create a grid of predictor values that includes a sequence ofages covering the observed range, all combinations of `Sex`, and all levels of `Pclass`.

```r
age_range <- range(titanic_clean$Age, na.rm = TRUE)
age_seq <- seq(from = age_range[1], to = age_range[2], by = 1)

sex_levels <- unique(titanic_clean$Sex)
pclass_levels <- sort(unique(titanic_clean$Pclass))

prediction_grid <- expand.grid(
  Age = age_seq,
  Sex = sex_levels,
  Pclass = pclass_levels
)

head(prediction_grid) # Short look
```

```
##    Age  Sex Pclass
## 1 0.42 male      1
## 2 1.42 male      1
## 3 2.42 male      1
## 4 3.42 male      1
## 5 4.42 male      1
## 6 5.42 male      1
```

2. Use the fitted model to compute the estimated survival probability for each combination of predictors.

```r
prediction_grid$predicted_prob <- predict(
  full_model, # glm
  newdata = prediction_grid,
  type = "response" # returns probabilities, not log-odds
)

head(prediction_grid)
```

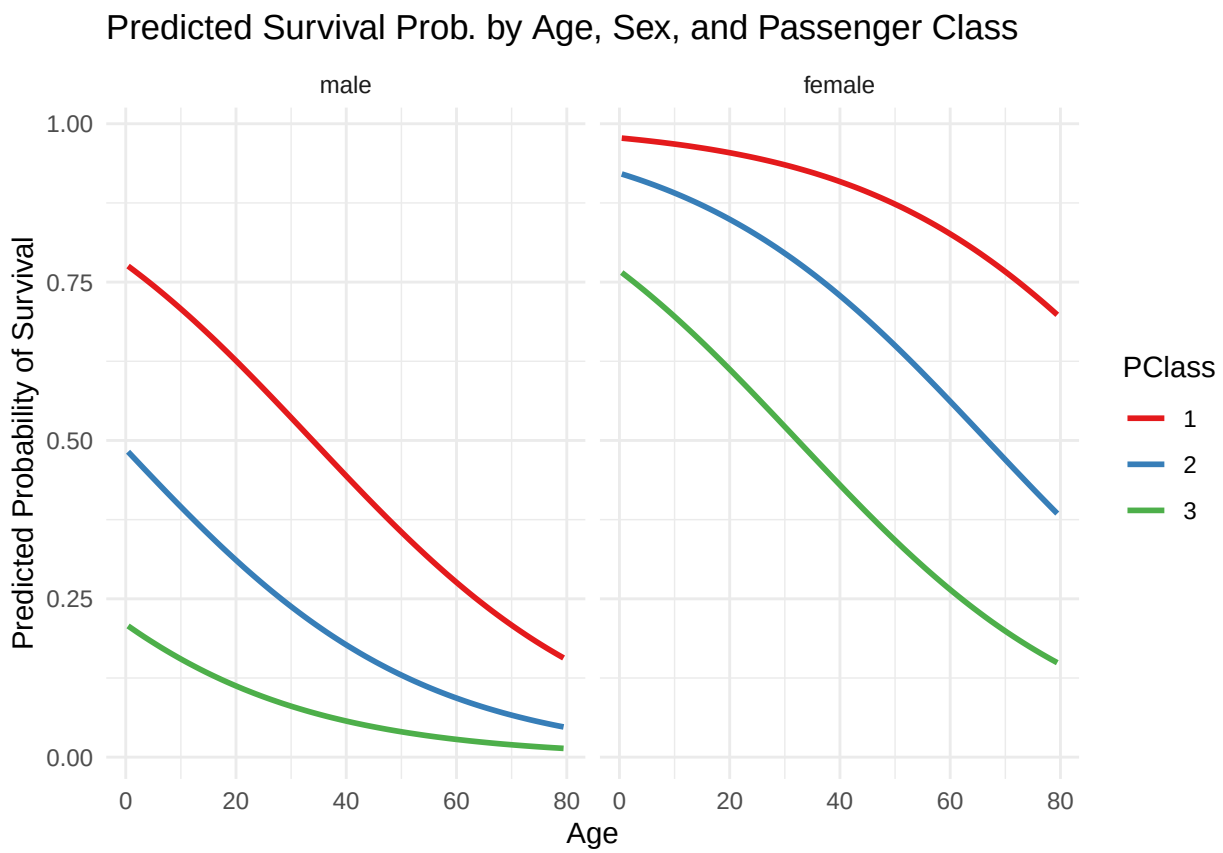
```
##    Age  Sex Pclass predicted_prob
## 1 0.42 male      1      0.7753373
```

```
## 2 1.42 male      1      0.7688293
## 3 2.42 male      1      0.7621906
## 4 3.42 male      1      0.7554220
## 5 4.42 male      1      0.7485242
## 6 5.42 male      1      0.7414985
```

3. Create a plot showing survival prob. on the y-axis,age on the x-axis, separate curves for different passenger classes, and faceting by sex.

```r
ggplot(prediction_grid, aes(x = Age, y = predicted_prob, color = factor(Pclass))) +
  geom_line(size = 1) +
  facet_wrap(~ Sex) +
  labs(
    title = "Predicted Survival Prob. by Age, Sex, and Passenger Class",
    x = "Age",
    y = "Predicted Probability of Survival",
    color = "PClass"
  ) +
  theme_minimal() +
  scale_color_brewer(palette = "Set1")
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



Passenger A: 25 yr-old, female, 1st class. **Passenger B:** 40 yr-old, male, 3rd class. **Passenger C:** 60 yr-old, female, 2nd class.

4. For each of the 3 passengers above estimate the prob. of survival using the fitted model, and report the estimated probability.

```
new_passengers <- data.frame(
  Age = c(25, 40, 60),
  Sex = c("female", "male", "female"),
  Pclass = factor(c(1, 3, 2), levels = levels(titanic_clean$Pclass))
)

predicted_probs <- predict(
  full_model,
  newdata = new_passengers,
  type = "response"
)

results = cbind(new_passengers, predicted_prob=predicted_probs)
results
```

```
##   Age    Sex Pclass predicted_prob
```

```
## 1  25 female      1     0.94544163
## 2  40   male      3     0.05701133
## 3  60 female      2     0.56170816
```

5. Construct a 95% CI for the survival probability of each passenger. Your CIs should be based on predictions on the log-odds scale,and transformations back to probability scale.

```r
# On the logit scale
pred_logit <- predict(
  full_model,
  newdata = new_passengers,
  type = "link",
  se.fit= TRUE
)

lower_CI <- pred_logit$fit - 1.96 * pred_logit$se.fit
upper_CI <- pred_logit$fit + 1.96 * pred_logit$se.fit

c(lower_CI, upper_CI)
```

```
##          1          2          3          1          2          3
##  2.3118971 -3.2571027 -0.3794717  3.3928649 -2.3545057  0.8756666
```

```r
inv_logit <- function(x) 1 / (1 + exp(-x))

predicted_prob <- inv_logit(pred_logit$fit)
lower_prob <- inv_logit(lower_CI)
upper_prob <- inv_logit(upper_CI)

results <- cbind(
  new_passengers,
  predicted_prob = predicted_prob,
  lower_95CI = lower_prob,
  upper_95CI = upper_prob
)

results
```

```
##   Age    Sex Pclass predicted_prob lower_95CI upper_95CI
## 1  25 female      1     0.94544163  0.9098576  0.9674808
## 2  40   male      3     0.05701133  0.0370725  0.0867083
## 3  60 female      2     0.56170816  0.4062543  0.7059234
```

6. Briefly explain why confidence intervals are constructed on the link scale rather than directly on the probability scale, and how uncertainty in the estimated coefficients affects prediction uncertainty.

CIs for predicted probability in logistic regression are constructed on the logit scale because the linear predictor $\eta = x^\top \hat{\beta}$ is approximately normally distributed, allowing standard errors and symmetric CIs to be computer.

Directly constructing CIs on the probability scale is problematic since the transformation from log-odds to probability is non-linear and the distribution of p is skewed, especially near 0 or 1. Transforming the interval back into the probability scale produces an asymmetric confidence interval that properly reflects this.

f) Interaction effects in logistic regression

Consider a model:

$$\log(\frac{p_i}{1-p_i}) = \beta_0 + \beta_1 \text{Age}_i + \text{Sex}_i + \text{Pclass}_i + (\text{Age} \times \text{Sex})_i + (\text{Age} \times \text{Pclass})_i$$

where $p_i = P(\text{Survived}_i = 1)$.

1. Write down the full interaction model explicitly, clearly indicating the reference levels for `Sex` and `Pclass`, the dummy variables created for each categorical predictor, and how interaction terms modify the slope of `Age`.

$$\log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 Age_i + \beta_2 Sexmale_i + \beta_3 Pclass2_i + beta_4 Pclass3_i$$
$$+\beta_5(Age_i \times Sexmale_i) + \beta_6(Age_i \times Pclass2_i) + \beta_7(Age_i \times Pclass3_i)$$

(This is also part of (3)):

β_0 is the log-odds of survival for the reference group. β_1 is slope of Age for the reference group. β_2 is difference in log-odds for male vs female at Age=0 (baseline). β_3, β_4 are differences in log-odds for Pclass 2 or 3 vs 1 at Age=0. β_5 is an interaction term representing additional change in slope of Age for males relative to females. β_6, β_7 are interaction terms representing additional change in slope of Age for Pclass 2 or 3 relative to Pclass1.

2. Fit the interaction model in R.

```r
titanic_clean$Sex <- factor(titanic_clean$Sex)
titanic_clean$Pclass <- factor(titanic_clean$Pclass)

# logistic regression with interactions
interaction_model <- glm(
  Survived ~ Age * Sex + Age * Pclass,
  data = titanic_clean,
  family = binomial
)

summary(interaction_model)
```

```
##
## Call:
## glm(formula = Survived ~ Age * Sex + Age * Pclass, family = binomial,
##     data = titanic_clean)
##
## Coefficients:
```

```
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.19239    0.67929   3.227 0.001249 **
## Age          0.02037    0.02026   1.006 0.314620
## Sexmale     -0.97328    0.46558  -2.090 0.036577 *
## Pclass2     -0.22002    0.74179  -0.297 0.766764
## Pclass3     -1.79719    0.65914  -2.727 0.006400 **
## Age:Sexmale -0.05923    0.01668  -3.552 0.000382 ***
## Age:Pclass2 -0.04026    0.02198  -1.831 0.067086 .
## Age:Pclass3 -0.02686    0.02024  -1.327 0.184420
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 964.52  on 713  degrees of freedom
## Residual deviance: 631.83  on 706  degrees of freedom
## AIC: 647.83
##
## Number of Fisher Scoring iterations: 5
```

3. Interpret Coefficients: explain what an interaction term represents on the log-odds scale and how the effect of age changes across different sexes and passenger class.

Explanation is in part 1!

Age & Sex: For females, slope is 0.02037, corresponding to survival slightly increasing with age. For males, slope is $0.02037 - 0.05923 =$ -0.0389; survival slightly decreases with age.

Age & Pclass: Pclass2 slope was $0.02037 - 0.04026 =$-0.0199, a slight decrease in survival with age. Pclass3 slope was $0.02037 - 0.02686 =$ -0.0065; nearly flat, very small effect on age.

4. Compare the interaction model to the additive model using a likelihood ratio test. State the null hypothesis, identify the full and reduced models, and report the test result.

The reduced model does not include the $Age \cdot Sex$ nor the $Age \cdot Pclass$ interaction terms present in the full model defined at the start of (f).

Null hypothesis: $\beta_4 = \beta_5 = 0$. Alternative: At least one interaction term is non-zero.

```r
additive_model <- glm(Survived ~ Age + Sex + Pclass,
                      data = titanic_clean,
                      family = binomial)

interaction_model <- glm(Survived ~ Age * Sex + Age * Pclass,
                         data = titanic_clean,
                         family = binomial)

# Likelihood ratio test
anova(additive_model, interaction_model, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Survived ~ Age + Sex + Pclass
```

```
## Model 2: Survived ~ Age * Sex + Age * Pclass
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       709     647.28
## 2       706     631.83  3    15.45 0.001469 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5. Based on the test and coefficient interpretations, assess whether interaction effects are supported by the data and comment on whether the interaction model provides a meaningful improvement over the additive model.

The p-value is 0.001469, below our 0.05 significance level. This gives us reason to reject the null hypothesis. We can now conclude that the interaction model *does* provide a statistically significant improvement over the additive model. Most of the improvement come from the `Age x Sex` interaction, while the `Age x Pclass` interactions are weaker. This indicates the effect of age on survival differs by sex and to a lesser extent by passenger class.